

Pratham Singhal

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EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi) <i>B.Tech in Computer Science and Design; CGPA: 8.00/10.00</i>	New Delhi, India 2021 – 2025
Bal Bharati Public School <i>CBSE – Class XII: 97.2% Class X: 98.4%</i>	New Delhi, India 2018 – 2020

EXPERIENCE

Software Development Engineer 1 (SDE1) <i>NatWest Group</i> - Owned and scaled a Java (Spring Boot) / Python REST API platform for international payments processing, adopted internally by 12+ engineering teams across NatWest for upstream and downstream banking integrations. - Engineered a webhook- and callback-driven e-signature integration with Adobe eSign handling 8,000+ requests/hour in production, secured with mutual TLS (mTLS) and password/fingerprint-based authentication. - Enforced authentication, rate limiting, and traffic policies for 15+ REST APIs via API gateways (Tyk, Apigee), standardizing contracts with Swagger/OpenAPI; architected persistence across MongoDB/Oracle SQL with a Redis/Valkey cache on AWS ElastiCache, cutting p95 latency $\sim 35\%$. Containerized services with Docker, deployed via Kubernetes, and owned CI/CD on AWS (S3, CloudWatch), cutting release time from hours to under 15 minutes. - Established asynchronous IBM MQ workflows and multithreaded async/sync Java services; led vulnerability/CVE remediation and developed internal Streamlit/invoke developer tooling.	July 2025 – Present <i>Gurugram, India</i>
Undergraduate Research Assistant [Code] [Paper] <i>IIIT-Delhi (Guide: Dr. Dhruv Kumar)</i> - Built a Python data pipeline (web scraping + PDF-to-CSV parsing) to extract raw NIRF ranking data from government PDF reports into structured datasets, powering a React-based transparent ranking dashboard. - Conducted user research (surveys and interviews) on India's opaque NIRF ranking methodology, applied HCI principles to redesign the ranking criteria and interface for usability, and co-authored the findings into a research paper.	Aug. 2023 – Apr. 2024 <i>New Delhi, India</i>

PROJECTS

Mr. Pasteur – Database-First Dairy Commerce System <i>Python, MySQL, SQL</i> [GitHub] - Designed a 3NF-normalized 16-table MySQL schema with 5 business-rule triggers enforcing row-level stock locking, cancellation windows, and discount caps directly at the schema level, not trusted to application code. - Implemented atomic checkout via a single ACID stored procedure and schema-level RBAC across 4 roles; reproduced and resolved real concurrency bugs (lost-update wallet race, last-unit stock race) with live transaction-conflict demos.
AI Repo Buddy – LLM-Assisted Repository Setup Tool <i>Python, Streamlit, Gemini API</i> [GitHub] - Created a Streamlit tool that clones any GitHub repository, detects its stack across 6+ ecosystems (Python, Node, Go, Rust, Java/Maven/Gradle, Docker Compose), and installs dependencies in an isolated per-clone virtual environment. - Orchestrated concurrent multi-service startup (frontend + backend) with live log streaming and auto port detection, plus self-healing retries that repair broken paths; integrated the Gemini API for README-based command extraction with CI-gated lint/tests.
BetterWorldNow – Environmental Engagement Platform <i>HTML, React, JavaScript, CSS</i> [GitHub] [Website] - Scoped the product with a 116-response survey and 11 field interviews, using a chi-squared test to validate that incentive-based rewards – not just awareness content – drive user action, and translated the finding into a points-and-rewards product loop. - Shipped the full-stack platform (HTML, React, JS, CSS) end-to-end: gamified quizzes, a leaderboard, a coupon-redemption rewards engine with real eco-brand partners, and a theme-aware, animation-rich UI (light/dark mode, custom SVG effects).
Hindi Text Summarization – NLP Pipeline <i>Python, PyTorch, HuggingFace, IndicBART</i> [GitHub] - Delivered an end-to-end Hindi abstractive summarization pipeline on the $\sim 70\text{K}$-article XL-Sum dataset, from data cleaning through two zero-training extractive baselines (Lead-N, TextRank) to a fine-tuned IndicBART transformer. - Evaluated outputs with ROUGE and BERTScore, engineered the pipeline to run reproducibly on CPU-only hardware, and packaged it as a tested Python library with CLI inference.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, Haskell, SQL, JavaScript, HTML/CSS
Machine Learning & Data: PyTorch, HuggingFace Transformers, NLP, NLTK, NumPy, Pandas, Matplotlib
Frameworks, Databases & Caching: Spring Boot, Flask, Streamlit, REST APIs, MongoDB, Oracle SQL, MySQL, Redis, Valkey
Cloud, DevOps & API Infrastructure: AWS (S3, ElastiCache, CloudWatch), Docker, Kubernetes, CI/CD, Git, Maven, Gradle, Swagger/OpenAPI, Tyk, Apigee, mTLS, IBM MQ, Rate Limiting
CS Fundamentals & System Design: Data Structures & Algorithms, Object-Oriented Design, System Design (HLD/LLD), Microservices Architecture, Operating Systems, Computer Networks, Multithreading & Concurrency, Distributed Systems, Scalability & Caching Strategies, Agile/Scrum